

Stage 1 Tutorial Activity

- Why Software Engineering Still Matters

Stage 1 | Introducing Software Technology Case Study with Python and Guided AI use

Learning goals

- Explain why software engineering is broader than coding.
- Identify stakeholders in a simple software problem.
- Recognise missing requirements.
- Critically evaluate AI-generated feature suggestions.
- Explain why AI output should not automatically be treated as correct.

Activity 1 - Think-Pair-Share (10 minutes)

If ChatGPT or Copilot can produce a 100-line Python application very quickly, what knowledge does a software engineer still need?

1. The ability to discover, clarify, and validate what real human clients and end-users actually need, while filtering out hallucinations, unnecessary features, and scope creep.
2. Understanding how components interact, ensuring software adheres to data privacy standards (e.g., patient health records), and designing systems that scale beyond a single script.
3. The capability to rigorously test edge cases, debug hidden logic errors, and assess whether generated code is safe, robust, and maintainable in a production environment.

Activity 2 - Is This Software Engineering? (10 minutes)

Scenario A: A student writes a 50-line Python calculator.

Scenario B: A team develops a payroll system used by 5,000 employees.

Scenario C: An AI assistant generates a simple appointment application from one prompt.

Scenario	Programming?	Software engineering?	Why?
A	Yes	No	It involves writing functional syntax and logic to solve a bounded mathematical problem, but it lacks formal lifecycle processes, external stakeholders, non-functional requirements, and ongoing maintenance
B	Yes	Yes	It requires coding as well as handling stakeholders, stringent

			financial/legal regulations, security, data persistence, concurrency, automated testing, and long-term maintainability.
C	Yes	No	Generating syntax from a prompt does not define verified user needs, establish architecture, perform integration testing, ensure medical data compliance, or manage client change requests.

Activity 3 - SmartCare Problem Analysis (20 minutes)

Client statement: SmartCare Community Clinic currently uses spreadsheets and paper records to manage patients and appointments. The clinic wants new software to improve these processes.

Task 1 - Identify stakeholders

Stakeholder	What do they need?
Clinic Receptionists / Admin Staff	A fast, reliable way to view GP availability, book or cancel patient appointments without double bookings, and quickly find patient contact records.
Healthcare Practitioners (GPs)	Clear visibility of their daily schedules, reliable patient history at consultation time, and prevention of schedule clashes.
Patients	Timely consultations, consistent appointment confirmations, and assurance that their personal health information is accurate and private.
Clinic Management	Basic operational reports (e.g., appointment volumes, cancellation rates) and a simple, manageable system that resolves operational bottlenecks without enterprise overhead.

Task 2 - Identify current problems

1. Inability to detect schedule overlaps across paper and spreadsheets.
2. Fragmented storage makes retrieving patient history slow and error-prone.
3. No reliable tracking for whether appointments are confirmed, completed, or cancelled.
4. Staff cannot easily see when individual GPs are working or free.

Task 3 - Ask client questions

1. What are the clinic's standard operating hours and the default duration for each appointment slot (e.g., 15 vs 30 minutes)?
2. How many GPs currently practice at the clinic, and do they work on fixed recurring schedules?

3. When an appointment is cancelled, should the record be completely removed or archived with a "Cancelled" status for reporting history?
4. What specific details must be recorded for each patient (e.g., Medicare number, phone number, date of birth) to uniquely identify them?
5. What specific metrics or summaries does clinic management expect in their basic operational reports (e.g., daily totals, GP patient counts)?

Activity 4 - Critique an AI Response (15 minutes)

An AI assistant suggests: appointment management; facial-recognition login; AI diagnosis recommendations; patient search; online payment; practitioner schedule view; insurance processing; automatic treatment-plan generation.

Suggestion	Client evidence?	In scope?	Decision
Appointment management		Yes	Accept
Facial recognition login		No	Reject
AI diagnosis recommendations		No	Reject
Patient search		Yes	Accept
Online payment		No	Reject
Practitioner schedule view		Yes	Accept
Insurance processing		No	Reject
Treatment-plan generation		No	Reject

Exit question

Write one activity that a software engineer must perform and that cannot safely be delegated entirely to AI.

Requirements negotiation and scope boundary enforcement with human clients. While AI can produce syntax and brainstorm features, only a human engineer can interview stakeholders, understand real operational and legal constraints, detect unstated business rules, and reject technically feasible yet counterproductive or dangerous scope additions.